



## Tight-binding modeling of excitonic response in van der Waals stacked 2D semiconductors

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Atomically thin transition metal dichalcogenides (TMDCs) exhibit fascinating excitonic properties which are of critical importance for potential applications in nanoscale optoelectronics. In particular, stacked TMDC layers without lattice mismatch allow modulation of their band structures and optoelectronic properties through manipulation of the constituent materials along the stacking direction. Current understanding of TMDC layer-by-layer structures is mainly based on optical measurements and DFT calculations. In this work, we use a phenomenological tight-binding model, combined with DFT calculations, to understand the measured layer-dependent excitonic response of WS<sub>2</sub>. This explicit and effective model can quantitatively predict the layer-dependent excitonic states and can also be used to study interlayer excitons (direct and indirect exciton states) in TMDC multilayer structures. In addition, we find that the temperature dependence of the A exciton emission energy of monolayer WS<sub>2</sub> can be well described with the Varshni formula, and that the emission intensity variation with temperature is associated with thermal redistribution of exciton population and increased non-radiative damping due to the enhanced electron-phonon interaction at elevated temperatures.

### Conceptual insights

The GW approximation and Bethe-Salpeter equation (BSE) methods are regarded as standard means to numerically predict excitonic effects in van der Waals (vdW) stacked two-dimensional (2D) transition metal dichalcogenides (TMDCs), and their results are highly consistent with optical measurements. Despite the success of GW and BSE, an intuitive model is needed to capture the essential physics of the exciton-photon interaction in these stacked vdW structures under various conditions. A common model used to study the layer-dependent shift of excitonic energy is the quantum-well model with fractional-dimensional space. Although this model can fit the excitonic layer-dependence well with some tuning parameters, it barely reflects the nature of TMDC vdW bonds. Here, we use a phenomenological tight-binding model, combined with DFT calculations, to understand the measured layer-dependent excitonic response of WS<sub>2</sub>. This simple yet effective model can quantitatively predict layer-dependent excitonic properties and be used to study interlayer excitons in TMDC heterostructures. In addition, we find that the temperature dependence of the A exciton emission energy of monolayer WS<sub>2</sub> can be well described with the Varshni formula, and the temperature-induced emission intensity variation is associated with thermal redistribution of the exciton population and increased non-radiative damping due to the enhanced electron-phonon interaction at elevated temperatures.

## Introduction

Transition metal dichalcogenides (TMDCs), with the generalized formula MX<sub>2</sub> where M = Mo, W, or Pt, and X = S, Se, or Te, have attracted a great deal of fundamental research and interest for technological applications. Two-dimensional (2D) MX<sub>2</sub> is formed as a sandwich structure with chalcogen atoms in two hexagonal

planes separated by a plane of metal atoms. In particular, monolayer MX<sub>2</sub> materials exhibit direct band gaps in the visible and near-infrared region, making them ideal candidates as complementary metal oxide semiconductors. Due to the atomic thickness of 2D TMDCs, the dielectric screening effect is significantly suppressed, and is more than one order of magnitude weaker than that of bulk GaAs, leading to a dominant excitonic response at room temperature. The strong electron-hole interactions in these materials have triggered numerous studies in the context of many-body physics.<sup>1–4</sup> In addition to the influence on electronic states of many-electron systems, their optical properties have also attracted great attention. For instance, the breaking of inversion symmetry in odd-layer TMDCs enables control of the valley polarization with circularly polarized light<sup>5–10</sup> and strong excitonic second-harmonic generation.<sup>11–16</sup> It has been clearly shown that the stacking of 2D materials with van der Waals

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(vdW) bonds has a prominent effect on excitonic emission. A vdW heterostructure, which is deterministically formed by stacking different types of 2D materials together, can also exhibit new properties, such as long-lived interlayer excitons,<sup>17,18</sup> tunneling transistors, and light-emitting tunneling diodes, which are all absent in their constituent layers. Recently, layer-dependent ferromagnetism in a vdW CrI<sub>3</sub> 2D material (ferromagnetism in the monolayer, anti-ferromagnetism in the bilayer, and then back to ferromagnetism in the trilayer), which is not affected by the substrate coupled with it, has been predicted<sup>19</sup> and observed.<sup>20,21</sup> Nowadays, numerical studies of exciton states with GW and Bethe–Salpeter equation (BSE) methods are regarded as a standard way to describe semiconductor excitonic effects, and their results are highly consistent with optical measurements of various 2D materials.<sup>22,23</sup> Despite the success of the GW and BSE methods in calculating exciton states in 2D materials, an intuitive model, which can capture the essential physics of 2D layer-by-layer structures, is needed to understand the physical mechanisms of light–exciton interactions in these structures under different conditions.

In this work, we characterize the number of layers in WS<sub>2</sub> through examining their breathing modes in the low wavevector region, which indicates the vdW nature between WS<sub>2</sub> layers.<sup>24</sup> We then examine the layer-number-dependent A and B excitons with differential reflection spectroscopy. The extracted layer-dependent exciton resonance energies can be well described with an intuitive tight-binding model, which demonstrates a more general interpretation of the layer-dependent optical properties of WS<sub>2</sub> than the quantum well model with fractional-dimensional space. Finally, we observe an obvious temperature-dependent polaron effect on the photoluminescence emission energy and intensity of monolayer WS<sub>2</sub>, consistent with the Varshni formula and the physical scenario of the thermal redistribution of the exciton population, which reveals the similarity between the excitonic states in WS<sub>2</sub> and the electronic levels of traditional semiconductors.

## Experimental

### Synthesis

Monolayer and few-layer WS<sub>2</sub> triangular flakes were synthesized on 0.35 mm thick sapphire (Al<sub>2</sub>O<sub>3</sub>) substrates by chemical vapor deposition (CVD) reaction of WO<sub>3</sub> and sulfur. A sapphire wafer was first diced into 1 × 1 cm<sup>2</sup> pieces, washed in an ultrasonic bath with acetone and isopropanol, and then blow dried with UHP nitrogen gas. Subsequently, WO<sub>3</sub> thin films were transferred into a quartz tube reactor for sulfuration for 60 min at 30 Pa and ~900 °C. A mixed Ar and H<sub>2</sub> flow (8:1 in volume flow rate) was used to convey the WO<sub>3-x</sub> vapor species to the downstream substrates, and S vapor was generated from S powders placed upstream from a lower temperature region (~100 °C) which was controlled independently.

### Characterization

Raman spectra were recorded under excitation at 488 and 633 nm to determine the thicknesses of the WS<sub>2</sub> flakes.

Photoluminescence spectra from excitation at 488 and 532 nm as well as differential reflectance spectra were measured to study the thickness-dependent excitonic energies. The 532 and 633 nm lasers were installed in a confocal optical microscope system (NT-MDT NTEGRA Spectra Solar), and the laser beams were focused on the sample surface by a 100× objective (N/A = 0.90). Optical spectra from excitation at 488 nm were obtained on a micro-Raman spectrometer (HORIBA HR800), and the laser beam was illuminated onto the sample through a 100× objective (N/A = 0.80). Differential reflectance is defined as  $\delta R = R_W/R_S - 1$ , where  $R_W$  and  $R_S$  are the reflected light intensities from the sapphire substrate with and without the WS<sub>2</sub> flakes, respectively.<sup>25,26</sup> In the differential reflectance measurements, broadband white light from a tungsten quartz halogen source was focused on the sample by a 100× objective (N/A = 0.80); reflected signals were collected with the same objective and delivered to a CCD camera (Pixis 400) equipped with a spectrometer (Princeton Instrument, SP-2300) for spectral analysis. In the Raman measurements, the on-focus beam radius was about 1 μm which is smaller than the typical size of each layer region (larger than 2 μm in general). In the reflectance measurement, we used a weakly focused white light with a typical beam spot size of more than 100 μm to illuminate the flakes, and the reflectance from each layer region was measured by putting a pin hole in the image plane to restrict the region for signal collection. The corresponding measurement region was also about 1 μm.

### Computational methods

Density functional theory (DFT) calculations based on the conventional Kohn–Sham Hamiltonian were performed using the Perdew–Burke–Ernzerhof (PBE) local density approximation (LDA).

## Results and discussion

Fig. 1(a) shows the optical photograph of a CVD-grown WS<sub>2</sub> island on a sapphire substrate, which exhibits monolayer, bilayer, multilayer, and bulk grains. The specific thicknesses (*i.e.*, layer numbers) of this flake are determined with confocal Raman measurements. Fig. 1(b) shows the corresponding Raman mapping of the A<sub>1g</sub> peak intensity at 633 nm excitation, which demonstrates significant variation in Raman intensity over different grains. Low-frequency Raman spectra for points 1, 2, 3, 4, 5 and 6 as denoted in Fig. 1(a) are presented in Fig. 1(c). One traditional means of confirming the layer numbers of WS<sub>2</sub> is the use of the high-frequency Raman modes, E<sub>2g</sub><sup>1</sup> and A<sub>1g</sub>, excited at a 488 nm wavelength. Differing from the high-frequency Raman modes, low-frequency Raman spectra disclose more remarkable layer dependence, as presented in Fig. 1(c). The Raman peaks at ~32 cm<sup>-1</sup> and ~18 cm<sup>-1</sup> observed for the bilayer grain undergo red shift as the layer number increases, and they appear at ~12 cm<sup>-1</sup> and ~7 cm<sup>-1</sup> for the 5-layer flake. In accordance with the previously observed softening trend of Raman modes in MoS<sub>2</sub> and MoSe<sub>2</sub> with increasing layer number,<sup>24,27,28</sup> we tentatively assign these two

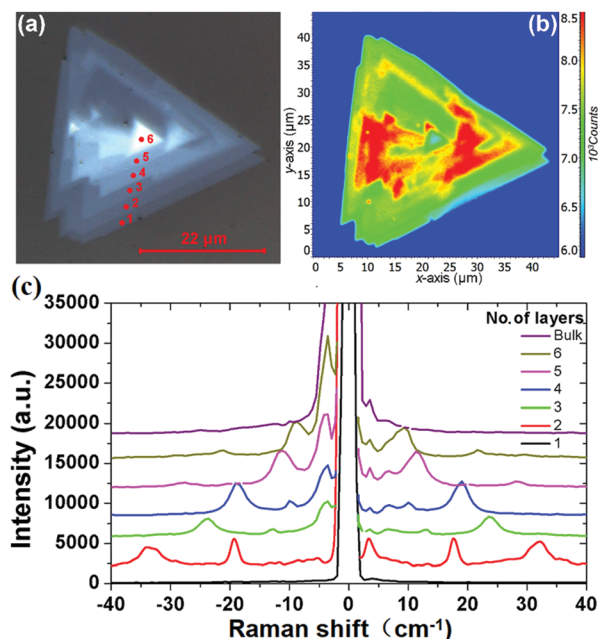


Fig. 1 (a) Bright-field optical image and (b) corresponding  $A_{1g}$  Raman intensity mapping of a  $WS_2$  island flake; (c) low-frequency Raman spectra of the  $WS_2$  island measured from points 1, 2, 3, 4, 5 and 6 as denoted in (a) under 488 nm excitation, which correspond to the 1L, 2L, 3L, 4L, 5L, 6L and the bulk  $WS_2$  regions.

Raman peaks to the  $B_1$  and  $B_2$  layer breathing modes,<sup>24</sup> which are closely related to the number of layers of  $WS_2$ .

Fig. 2(a) shows the differential reflectance spectra of a  $WS_2$  flake with different layer numbers. Two absorption peaks from the monolayer region can be observed at  $\sim 1.98$  and  $\sim 2.36$  eV, which correspond to the A and B excitonic energies, respectively. They arise from the direct gap transitions at the  $K$  ( $K'$ ) points of the first Brillouin zone.<sup>29</sup> As the layer number increases, both A and B excitons gradually shift to lower energies. For the bulk  $WS_2$  region, the A and B excitonic peaks shift to  $\sim 1.86$  and  $\sim 2.23$  eV, respectively.

Fig. 2(b) summarizes the peak positions of A and B excitons. It seems that the energy difference between the A and B peaks remains approximately 400 meV for all layers, in accordance with the spin-orbit splitting energy in the valence band of  $WS_2$ .<sup>30</sup>

A common model used to study the layer-dependent shifts of excitonic energies in 2D TMDCs is the quantum well model with fractional-dimensional space.<sup>31</sup> Although this model can fit the layer-number-dependent excitonic resonance energies well with some tuning parameters, it cannot reflect the nature of the vdW interlayers in which the potential energy is inhomogeneous across the structure. By making good use of the weak vdW bonds between layers, we can use single-layer valence and conduction bands to span the basis of the Hamiltonian of a vdW interlayer structure. The tight-binding model is applicable to vdW interlayers with finite extension of the wave function between interlayer atoms. This model has been widely used to calculate the band structure of graphene vdW interlayers.<sup>31–33</sup> Under these conditions, similar to the case of the conduction band of black phosphor vdW interlayers,<sup>36</sup> here we employ a

phenomenological tight-binding model used for open-ended linear chain systems to model the TMDC vdW interlayer structures:

$$H_0 = \pm \sum_{j=1}^N \frac{E_{go}}{2} C_j^\dagger C_j + \left( \sum_{j=1}^{N-1} \gamma^{(s)} C_{j+1}^\dagger C_j + \text{h.c.} \right) \quad (1)$$

where  $E_{go}$  is the bandgap of the single-layer  $WS_2$ , and  $\gamma^{(s)}$  is the interlayer hopping constant. As a result, the corresponding optical band gap ( $E_g$ ) can be determined using the open-ended tight-binding model:

$$E_g = E_{go} - 2 \left( \gamma^{(c)} + \gamma^{(v)} \right) \text{Max} \left[ \cos \left( \frac{n}{N+1} \pi \right) \right] \quad (2)$$

The allowed values of  $n$  are 1, 2, 3, ...,  $N$ . We further extend this model to study the excitonic effects in TMDCs. In the framework of this model, the  $N$ -layer  $WS_2$  is equivalent to a finite 1D lattice of  $N$  identical atoms with only nearest-neighbour coupling. The Hamiltonian of an exciton envelope function,  $\phi(r)$ , formed in a vdW interlayer structure is given by:

$$\left[ \frac{-\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{\epsilon r} \right] \phi(r) = (E_{ex} - E_g) \phi(r) \quad (3)$$

where  $1/\mu = 1/m_h + 1/m_v$  ( $m_h$  and  $m_v$  are the effective masses of the hole and electron at the valence and conduction bands, respectively),  $(E_{ex} - E_g) = E$  is the negative of the exciton binding energy and  $E = -R / \left( m + \frac{1}{2} \right)^2$  is for a 2D system. Therefore,

the corresponding transition energy of the exciton at the  $K$  or  $K'$  valley (neglecting interlayer electron-hole interactions and layer-number-dependent dielectric screening effects) can be expressed as:

$$E_{ex} = E_{go} - 2 \left( \gamma^{(c)} + \gamma^{(v)} \right) \text{Max} \left[ \cos \left( \frac{n}{N+1} \pi \right) \right] - \frac{R_y}{\left[ m + \frac{1}{2} \right]^2} \quad (4)$$

where  $\gamma^{(c)} + \gamma^{(v)} = 0.842$  eV (determined by fitting the experimental results as shown in Fig. 2(b)), and the integers  $m$ ,  $n$  and  $N$  are the exciton quantum number, the interlayer state index and the number of layers, respectively, and  $R_y$  is the Rydberg constant for the  $WS_2$  exciton. The layer-number-dependent characteristics in eqn (4) are valid for  $n \geq 2$  and the change in optical bandgap from  $n = 1$  to 2 involves a direct to indirect bandgap transition (this issue will be discussed in a later section).

One can see from eqn (4) that the larger the layer number is, the lower the energy of the excitonic transition becomes, which is in agreement with the experimental results summarized in Fig. 2(b). Due to the localization of the exciton wavefunction, the ground state of the  $WS_2$  excitons (such as A and B excitons) can be approximated as the 1s state of a 2D exciton.<sup>22</sup> However, an additional modification is needed to study the layer-dependent dielectric function for  $m > 1$ .<sup>22,23</sup> Besides the reflectance spectra, the photoluminescence spectra shown in Fig. 3(b) (for points 1, 2, 3, 4, 5 and 6 as denoted in Fig. 1(a)) also

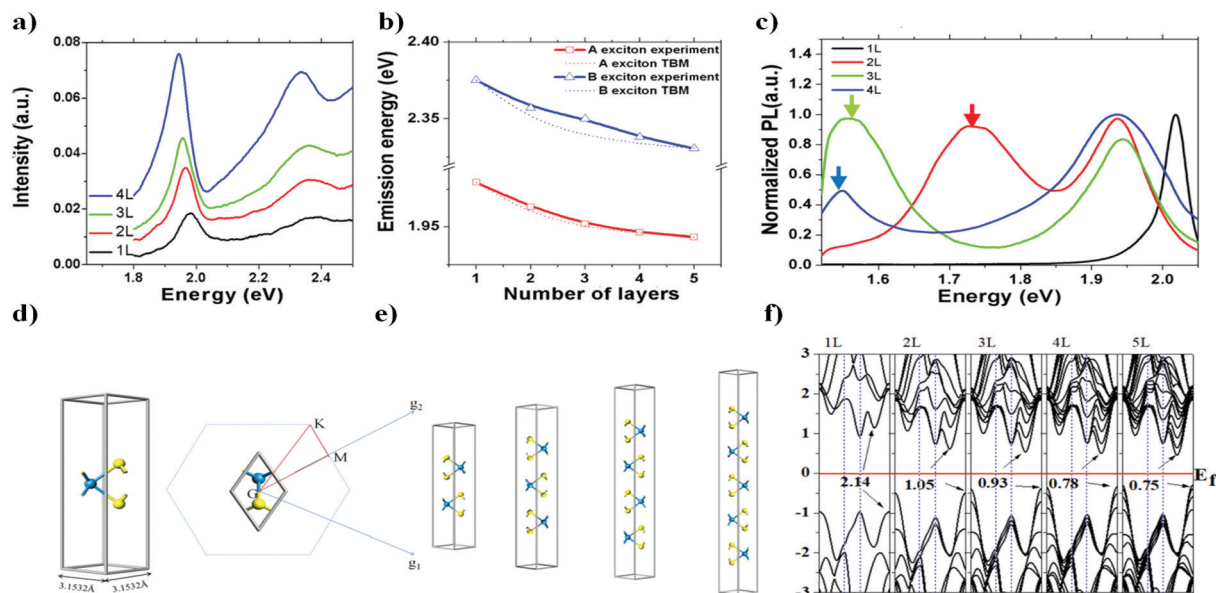


Fig. 2 (a) Differential reflectance spectra of a  $\text{WS}_2$  island flake with different layer numbers; (b) measured peak positions of A and B 1s state excitons with fitting curves from eqn (3). The symbols are experimental data, and the dotted lines are the fitting results using eqn (4); (c) photoluminescence spectra of the same flake under 488 nm excitation; (d) and (e) are atomic structures of  $\text{WS}_2$  for the DFT calculations; (f) calculated layer-number-dependent energy band structures, where the vertical axis is the energy in eV.

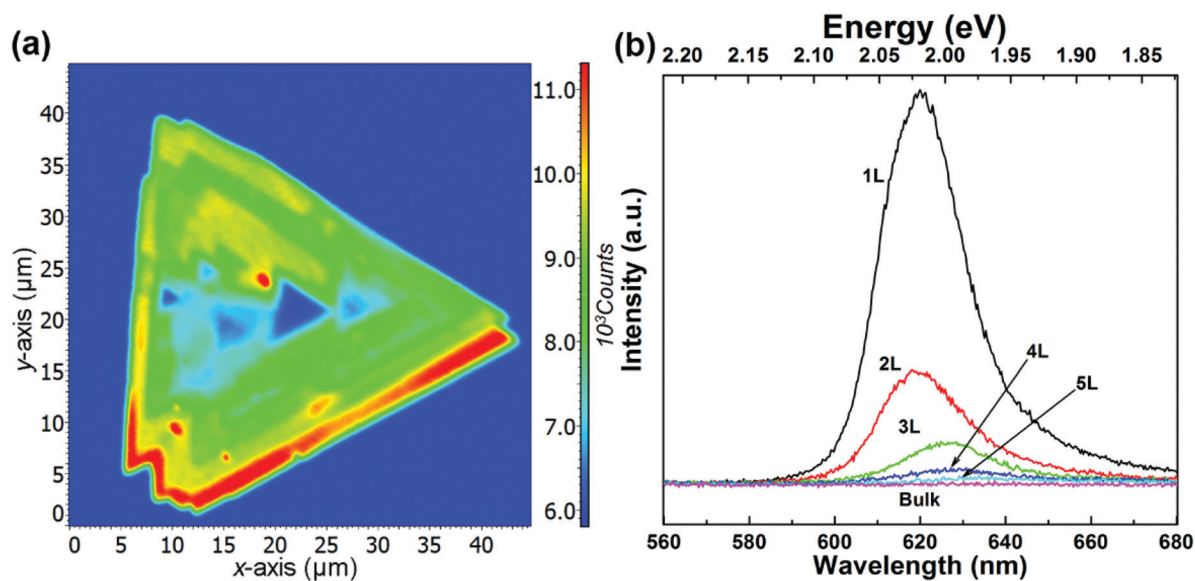


Fig. 3 (a) Fluorescence intensity map and (b) fluorescence spectra of  $\text{WS}_2$  films under 532 nm excitation.

exhibit significant layer-number dependence. Fig. 3(a) presents the corresponding emission intensity mapping of the  $\text{WS}_2$  island flake at 532 nm excitation. One can see that the PL signal intensity for  $n > 1$  is much lower than that for  $n = 1$  and the bulk  $\text{WS}_2$  region shows nearly no signal, which is due to the indirect bandgap nature. When an electron is excited to a higher-energy state under light illumination, it will relax to the minimum energy state *via* multi-phonon emission. For an indirect bandgap transition, electron-phonon interaction is required for conservation of crystal momentum. For radiative

recombination to occur in an indirect band gap material, this process must also involve the absorption or emission of a phonon with a momentum equal to the crystal momentum between the electron and the hole, which is less likely to occur. This explains why PL in indirect band gap materials is much lower than in the direct band-gap transition ones. As the  $\text{WS}_2$  flake thins to a few layers, the emission intensity from direct interband transitions dramatically increases and reaches the maximum at the monolayer limit. The emission peak also shifts from  $\sim 633$  nm for the 5L  $\text{WS}_2$  to  $\sim 620$  nm for the monolayer,



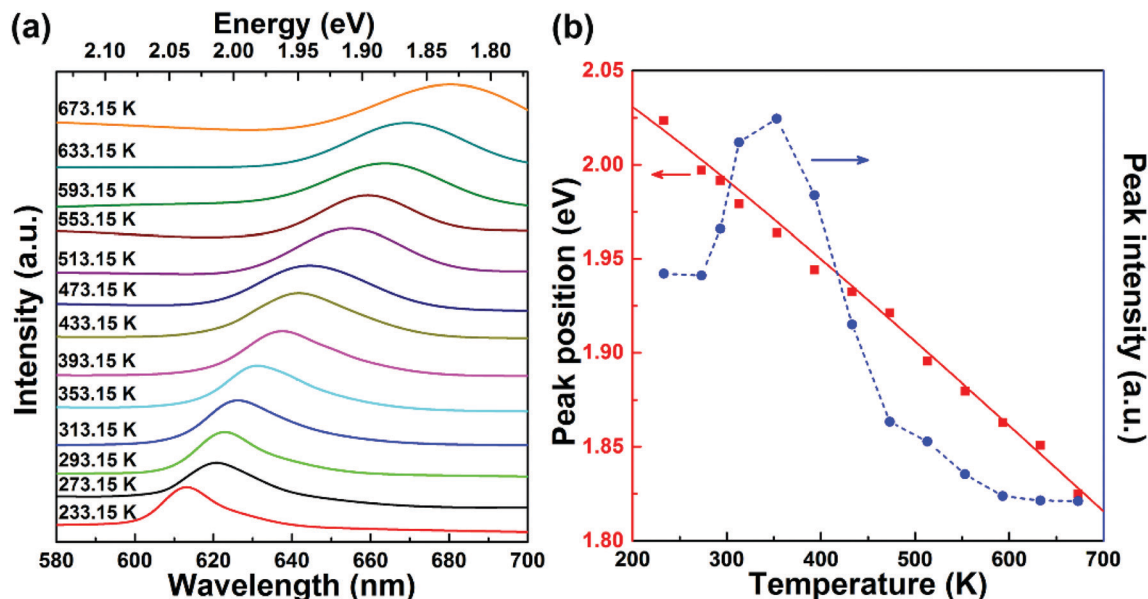


Fig. 4 (a) Temperature-dependent photoluminescence of monolayer WS<sub>2</sub>. The spectra are vertically offset from each other for clarity. (b) Emission peak position and intensity extracted from the photoluminescence spectra in (a). The solid line is the fit using the Varshni formula.

indicating the respective direct bandgap and indirect bandgap natures of monolayer and multilayer WS<sub>2</sub>.<sup>26,36–39</sup> In theory, our DFT results show that the bandgap transitions from direct to indirect when the layer number changes from monolayer to multilayer, which is different from in 2D black phosphorene.<sup>32</sup> Therefore, the decrease in both exciton energy and optical absorption per layer from mono- to bilayer is caused by the transition from an direct to indirect bandgap<sup>26,34</sup> rather than by fractional dimensions or a drastic change in dielectric constant in multilayer samples.<sup>35</sup> The validity of the tight-binding model can be checked with *ab initio* GW-BSE calculations, which explicitly include screening from the substrate and between the layers as well as self-energy and electron-hole interaction effects.<sup>36</sup> The extraordinary layer-dependent electronic structure of WS<sub>2</sub> vdW structures is caused by the spreading of electron wavefunction between WS<sub>2</sub> layers. Fig. 4(a) presents the photoluminescence spectra of monolayer WS<sub>2</sub> measured at different temperatures under 488 nm excitation. In order to clearly observe the shift of emission peaks, the spectra are vertically shifted from each other. Fig. 4(b) shows the temperature-dependent emission peak energies extracted from Fig. 4(a), and demonstrates a continuous redshift of emission energy with increasing temperature. Here the temperature-induced emission energy shift can be fitted with the Varshni formula:

$$E(T) = E_{\text{go}} - \alpha T^2 / (T + \beta) - R_y / (m + 1/2)^2 \quad (5)$$

where  $E_{\text{go}}$  is the bandgap at 0 K and  $\alpha$  and  $\beta$  are phenomenological fitting parameters.<sup>40</sup> By using eqn (5) to fit the emission peak positions in Fig. 4(b) and setting  $m = 0$  for the A exciton emission, we obtain  $E_{\text{go}} - R_y / (0 + 1/2)^2 = 2.08$  eV,  $\alpha = 4.8 \times 10^{-4}$  eV K<sup>-1</sup> and  $\beta = 190$  K. These parameters are comparable to those reported for excitonic emissions at low temperatures.<sup>41</sup> Also, one can see from Fig. 4(b) that the A exciton emission

intensity reaches the maximum as the temperature increases to 353.15 K, and then gradually decreases with further increases in temperature. The initial intensity rise may originate from the redistribution of exciton population from the underlying dark exciton states to the bright A exciton states.<sup>1,42</sup> Once the dark exciton redistribution is saturated at fairly high temperatures, the A exciton emission may begin to suffer from larger non-radiative damping at elevated temperatures, leading to the sharp decrease in the emission intensity. In fact, both the lowering of emission intensity and broadening of emission line width with increasing temperature are expected to be accompanied by a common effect known as line broadening, which is due to the enhanced electron-phonon interaction at high temperatures. Nevertheless, the good agreement of the Varshni formula and the measured WS<sub>2</sub> excitonic emission energies illustrates the close connection between the excitonic response in WS<sub>2</sub> and the band edge tendency in traditional semiconductors. This result may originate from the shifting of the band edge of WS<sub>2</sub> to maintain a constant binding energy as shown in eqn (4).

## Conclusions

We have examined the layer-dependent excitonic response of monolayer and vdW multilayer WS<sub>2</sub> through combining Raman, reflectance and photoluminescence imaging and spectroscopic measurements. We have observed a clear Raman breathing mode in the low-frequency Raman spectra of multilayer WS<sub>2</sub>, whose Raman shift allows rapid and unambiguous identification of the number of WS<sub>2</sub> layers. We have used a tight-binding model, combined with DFT calculations, to semi-analytically understand the layer-dependent excitonic states in multilayer WS<sub>2</sub>. This

simple yet effective model can quantitatively predict the layer-dependent excitonic response of WS<sub>2</sub>. We believe that an extension of this model could also be used to study interlayer excitons in general TMDC multilayer structures as well as other 2D semiconductors within the same theoretical framework. Finally, we found that the temperature-dependent emission energies of monolayer WS<sub>2</sub> could be well described with the Varshni formula, indicating that the temperature-induced redshift in emission energy comes from the shifting of the band edge in order to maintain a constant binding energy.

## Conflicts of interest

There are no conflicts to declare.

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